BEFORE YOU READ

### How smooth is the derivative?

Functions sort into classes by how well their derivatives behave. Each class sits inside the one around it.

$$C^\infty \subset C^1 \subset D \subset C^0$$

$C^0$ **Continuous** — $f$ may not exist — `|x|`

$D$ **Differentiable** — $f'$ exists everywhere but may be discontinuous — `x² sin(1/x), f(0) = 0`

$C^1$ **Continuously differentiable** — $f'$ exists and is continuous — `polynomials, sin x, eˣ`

$C^\infty$ **Smooth** — all higher derivatives exist and are continuous

TAKEAWAY

### Differentiable is not the same as $C^1$

A derivative can exist everywhere and still be discontinuous, as $x \sin(1/x)$ shows; by Darboux's theorem it can never have a jump.